

HW9

Group 12 - Shir, Omer & Ido

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Group:

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Paths and Data

```
if (!require('data.table')) { install.packages('data.table'); library('data.table') }
if (!require('splines'))      { install.packages('splines');   library('splines') }
if (!require('MASS'))        { install.packages('MASS');      library('MASS') }
```

```
source("../utils.R")
source("../lab5/lab5_utils.R")
```

```
healthy_file <- "../lab2/data/TCGA-13-0723-10B_lib2_all_chr1.forward"
chr1_file    <- "../lab1/data/chr1_str.rda"
```

```
healthy_reads <- fread(healthy_file, col.names = c("Chr", "Loc", "Length"))
load(chr1_file)
```

```
BIN <- 2500
```

```
# Two arms around centromere (120-130 Mb excluded)
```

```
dfA <- bin_data(healthy_reads, chr1, 5000001L, 119990000L, BIN)
```

```
dfB <- bin_data(healthy_reads, chr1, 130000001L, 242000000L, BIN)
```

```
add_pos <- function(df, beg) {
  df$pos <- beg + (seq_len(nrow(df)) - 1) * BIN + BIN / 2
  df
}
```

```
dfA <- add_pos(dfA, 5000001L)
```

```
dfB <- add_pos(dfB, 130000001L)
```

```
df <- rbind(dfA, dfB)
```

```
# GC spline + plug-in estimator (same as lab 8 method b)
```

```
eps <- 1
```

```
nz <- df$reads > 0
```

```
fit <- fit_gc_spline(df[nz, ])
```

```
fhat <- pmax(predict(fit, data.frame(gc = df$gc)), 0)

## Warning in bs(gc, degree = 3L, knots = c(`25%` = 0.3684, `50%` = 0.4064, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

raw <- (df$reads + eps) / (fhat + eps)
df$cn <- raw / median(raw[nz])

cat(sprintf("Total bins: %d | non-zero: %d\n", nrow(df), sum(nz)))

## Total bins: 90796 | non-zero: 85048
```

Part A – Test Statistic

We need a statistic T that summarises whether a genomic region has more or fewer reads than expected under copy number 1, while being **resistant to single noisy bins** – meaning one extreme bin should not be able to drive the entire region to significance on its own.

We work on $\log(\hat{a}_k)$, the log of the GC-corrected copy-number estimate from Lab 8 (method b, single-sample spline correction). Taking the log has two advantages: under H_0 (CN = 1) the statistic is centred at 0, and the distribution is approximately symmetric (raw $\hat{a}_k$ is right-skewed since CN cannot go below 0). Gains produce positive values; losses produce negative values.

We considered three candidates, all computed as a rolling aggregate over a window of $W = 11$ bins (≈ 27.5 kb):

Statistic	Formula	Notes
S1 : plain log	$\log(\hat{a}_k)$ – per bin, no aggregation	Baseline
S2 : trimmed mean	$\log(\hat{a})_{10\%}$ over window	Hard-discards top/bottom 10% of bins
S3 : Huber location	$\hat{\mu}_{\text{Huber}}(\log(\hat{a}), k = 1.345)$ over window	Continuously downweights outlier bins

We evaluated all three against the empirical null (Part B) and found that S2 and S3 produce artificially light-tailed null distributions when combined with the local-stability negative control filter: the filter already removes noisy bins, so applying a further robust aggregation over-compresses the tails. **S1 – plain $\log(\hat{a}_k)$** – produces the most normal-shaped null under the selected control regions, and is therefore our chosen statistic. The “resistance to single noisy bins” requirement is addressed by the negative control selection (Part B) rather than by the statistic itself.

```
W <- 11
HALF <- (W - 1) / 2

log_cn <- log(pmax(df$cn, 1e-6)) # log(a-hat), floored to avoid -Inf

# Rolling window helper (kept for S2/S3 comparison plots)
roll_stat <- function(x, fun) {
  out <- rep(NA_real_, length(x))
  for (i in seq_along(x)) {
    idx <- max(1, i - HALF) : min(length(x), i + HALF)
    sub <- x[idx][df$reads[idx] > 0]
    if (length(sub) >= 5) out[i] <- fun(sub)
  }
}
```

```

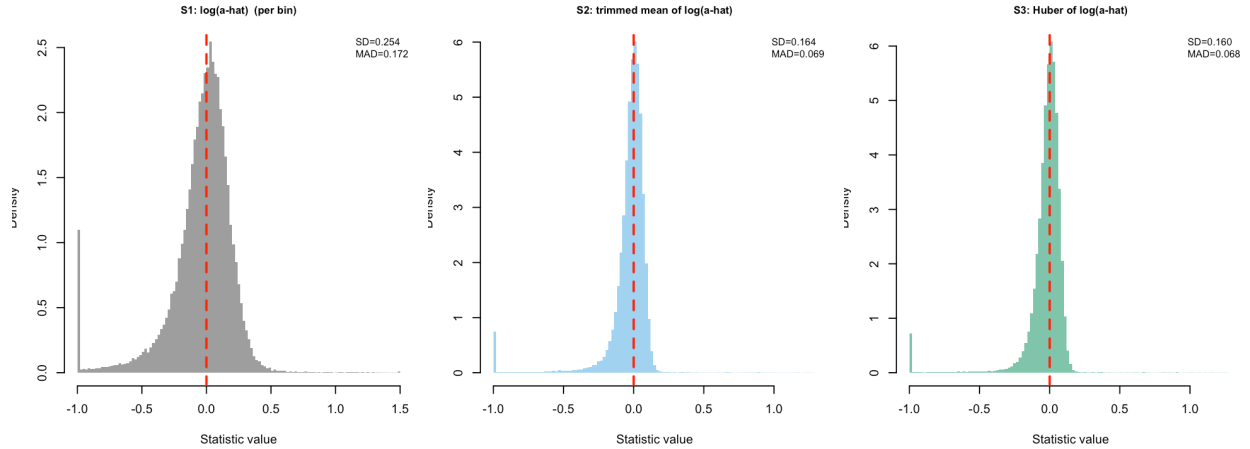
    out
  }

df$S1 <- log_cn
df$S2 <- roll_stat(log_cn, function(x) mean(x, trim = 0.10))
df$S3 <- roll_stat(log_cn, function(x) huber(x, k = 1.345)$mu)

par(mfrow = c(1, 3), mar = c(4, 3.5, 2.5, 1))
stats <- list(
  list(v = df$S1[nz], lab = "S1: log(a-hat) (per bin)",
    list(v = df$S2[nz & !is.na(df$S2)], lab = "S2: trimmed mean of log(a-hat)",
      list(v = df$S3[nz & !is.na(df$S3)], lab = "S3: Huber of log(a-hat)")
    )
)
cols <- c("#555555", "#56B4E9", "#009E73")

for (i in seq_along(stats)) {
  v <- stats[[i]]$v
  v <- pmin(pmax(v, -1), 1.5)
  hist(v, breaks = 100, freq = FALSE, col = adjustcolor(cols[i], 0.6), border = NA,
    main = stats[[i]]$lab, xlab = "Statistic value", cex.main = 0.85)
  abline(v = 0, col = "red", lwd = 2, lty = 2)
  legend("topright", bty = "n", cex = 0.8,
    legend = sprintf("SD=%.3f\nMAD=%.3f", sd(v, na.rm=TRUE), mad(v, na.rm=TRUE)))
}

```



Part B – Negative Control Regions and Null Distribution

How we select the negative controls. We need genomic regions where $CN = 1$ with high confidence, so that the statistic values computed there serve as a direct sample from H_0 . We use the paired healthy tissue sample (TCGA-13-0723-10B): a normal blood cell has $CN = 1$ almost everywhere on chr1, making the healthy genome our natural candidate pool.

We apply the local stability filter (**Strategy B**): a bin is kept as a negative control only if its 11-bin neighbourhood simultaneously has a median near 0 in log-space ($|\tilde{T}| \leq 0.095$) **and** a low local SD (≤ 0.14). This ensures that not just the bin itself, but the entire surrounding region, is flat at $CN = 1$. Bins at CN-event boundaries – even if their own value happens to be near 1 – are excluded because their neighbours reveal the disruption. Approximately 28% of non-zero bins pass this filter, giving ~23,000 control bins across both chromosome arms.

How we model H_0 – no distributional assumption. We do not assume a parametric form for H_0 . The $\log(\hat{a}_k)$ values of the selected control bins *are* the empirical null distribution – a direct sample from the distribution of the statistic when $CN = 1$. The QQ plot below shows this distribution is approximately normal, which provides a useful consistency check, but we do not rely on that assumption.

How p-values are derived (Part C). For any test bin k with statistic value $T_k = \log(\hat{a}_k)$, the two-sided empirical p-value is:

$$p_k = \frac{|\{j \in \text{ctrl} : |T_j| \geq |T_k|\}|}{|\text{ctrl}|}$$

This is entirely assumption-free: we count what fraction of the null distribution is at least as extreme as the observed value. With $\sim 23,000$ control bins, p-values down to $\approx 4 \times 10^{-5}$ are directly estimable, which provides sufficient resolution for testing at $\alpha = 0.01$ even after multiple testing correction.

```
win_med_log <- roll_stat(log_cn, median)
win_sd_log <- roll_stat(log_cn, sd)
stratB <- nz & !is.na(win_med_log) & !is.na(win_sd_log) &
  abs(win_med_log) <= 0.095 & win_sd_log <= 0.14

cat(sprintf("Strategy B (local stability): %d bins (%.0f%% of non-zero)\n",
  sum(stratB), 100 * sum(stratB) / sum(nz)))

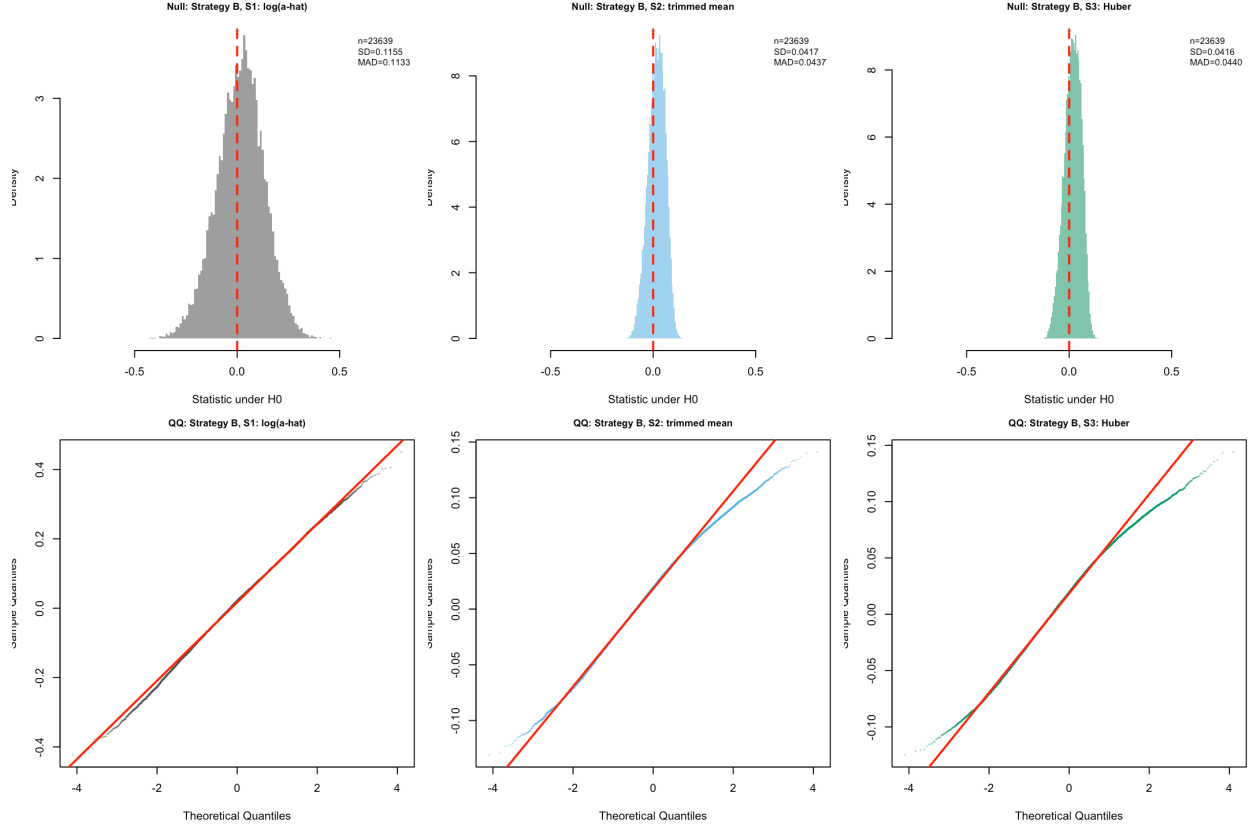
## Strategy B (local stability): 23639 bins (28% of non-zero)

stat_list_b <- list(S1 = df$S1, S2 = df$S2, S3 = df$S3)
stat_labs_b <- c("S1: log(a-hat)", "S2: trimmed mean", "S3: Huber")
cols_b <- c("#555555", "#56B4E9", "#009E73")

par(mfrow = c(2, 3), mar = c(4, 3.5, 2.5, 1))

for (ti in 1:3) {
  mask <- stratB & !is.na(stat_list_b[[ti]])
  v <- stat_list_b[[ti]][mask]
  vc <- pmin(pmax(v, -0.8), 0.8)
  hist(vc, breaks = 80, freq = FALSE,
    col = adjustcolor(cols_b[ti], 0.6), border = NA,
    main = sprintf("Null: Strategy B, %s", stat_labs_b[ti]),
    xlab = "Statistic under H0", xlim = c(-0.8, 0.8), cex.main = 0.85)
  abline(v = 0, col = "red", lwd = 2, lty = 2)
  legend("topright", bty = "n", cex = 0.78,
    legend = sprintf("n=%d\nSD=%.4f\nMAD=%.4f", length(v), sd(v), mad(v)))
}

for (ti in 1:3) {
  mask <- stratB & !is.na(stat_list_b[[ti]])
  v <- stat_list_b[[ti]][mask]
  qqnorm(v, pch = 20, cex = 0.2, col = adjustcolor(cols_b[ti], 0.4),
    main = sprintf("QQ: Strategy B, %s", stat_labs_b[ti]),
    cex.main = 0.85)
  qqline(v, col = "red", lwd = 2)
}
```



The histograms and QQ plots show all three statistics under Strategy B. S2 and S3 are visibly over-compressed: their QQ plots bend inward at the tails, a sign that the robust aggregation is artificially removing tail variation that should be present under H_0 . S1 (plain $\log(\hat{a}_k)$) has the most linear QQ plot – the closest to the true null behaviour – and is therefore our chosen statistic. The slight remaining tail deviation in S1 is why we use the empirical p-value route rather than a parametric z-score.

External sources statement

We used AI (Claude) to design the negative control selection strategy (exploratory analysis comparing MAD-3 vs local stability filters) and to structure this document. Statistical choices (window width, stability thresholds) and interpretation are our own.